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Predicting lake water quality index with sensitivity-uncertainty analysis using deep learning algorithms

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ABSTRACT

Regular monitoring and assessment of water quality is essential to maintain the quality of lake water. A commonly used method for assessing water quality is the Water Quality Index (WQI). Artificial intelligence (AI) can predict the WQI rapidly and more accurately than conventional methods. In this study, a stacking ensemble model based on Deep Learning (DL) was proposed to predict the WQI by integrating three models: Gradient Boosting Machine (GBM), Generalized Linear Model (GLM) and Neural Network (NN). The performance of this model was compared with that of Deep Dense Neural Network (DNN) and Convolutional Neural Network (CNN). The inclusion of a DNN model, which was used for the first time in water pollution research to perform sensitivity and uncertainty analysis in predicting WQI, added a new dimension to the workflow. The mean square error (MSE), mean absolute error (MAE), root mean square error (RMSE) and correlation coefficient (R^2) were used to evaluate the performance of the WQI prediction models. The results showed that the WQI of the water samples ranged from 90.75 to 145.29, indicating poor to very poor water quality. The validation results showed that the stack model was superior to all other models in terms of prediction accuracy, with MSE of 25.77, RMSE of 5.07, MAE of 3.5 and R^2 of 0.98. The DNN-based sensitivity analysis showed that pH and turbidity significantly affect the WQI and should be monitored to minimise water pollution. The uncertainty analysis indicated that electrical conductivity and total dissolved solids have the highest uncertainty in predicting WQI. This study provides researchers, decision makers and water scientists with accurate information on water quality and enables the implementation of clean production processes to reduce pollution and improve water quality.

1. Introduction

Due to population growth and unplanned development, natural water bodies such as lakes, ponds, wetlands, and rivers are facing threats such as pollution, encroachment, displacement, eutrophication, and overdevelopment (Cosgrove and Loucks, 2015; Prasad et al., 2022). The main cause of water pollution in lakes and ponds is untreated wastewater from domestic or industrial sources (Baghanam et al., 2020; Heneash et al., 2021), while other processes, such as leakage of CO_2 sequestration, can also contribute (Sun et al., 2022). Water pollution not only affects ecosystems but also has negative impacts on human, plant, and animal health (Wear et al., 2021). Thus, water quality assessment is crucial for water resources engineering and environmental management systems (Ek and Persson, 2020; Wuijts et al., 2021; Xiang et al., 2021; Prasad et al., 2022; Wan et al., 2022). However, integrating different physical and chemical parameters of water for assessment is challenging and requires sophisticated numerical models (Uddin et al., 2021).

The WQI integrates physico-chemical parameters of water into a single variable to describe the quality of water at a specific location and time (Zhao et al., 2021; Uddin et al., 2021). The WQI uses a single index value based on the ranking and weighting determined by expert opinion (Mallik et al., 2021). While various WQI approaches exist, such as the United States National Sanitation Foundation Water Quality Index (NSFWQI), the Florida Stream Water Quality Index (FWQI), the British

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Columbia Water Quality Index (BCWQI), the Canadian Water Quality Index (CWQI), and the National Water Quality Standards for Malaysia (INWQS) (Bui et al., 2020), the most widely accepted approach is the one developed by Horton (1965). However, traditional WQI methods have site-specific uncertainties and a complex and lengthy calculation process.

To overcome these limitations, researchers have combined WQI with AI to analyze and predict water quality (Uddin et al., 2021; Wang et al., 2019; Pandey and Verma, 2022). AI offers several advantages for water quality assessment, such as the ability to handle complex and non-linear structures, larger datasets at different scales, and the production of WQI values in real-time (Bui et al., 2020).

Initially, water quality analysis and prediction relied on mechanistic models, which constructed the water quality model by examining the mechanisms of water bodies. However, later, data-driven models emerged that were based on multiple datasets and statistical experiments for analyzing water quality in various bodies of water such as lakes, ponds, rivers, and estuaries (Hipsey et al., 2020; Wang et al., 2019). The earliest data-driven models purely utilized statistical techniques such as frequency ratio, fuzzy logic, and evidential-belief function (Raman et al., 2009; Manap et al., 2014). These statistics-based data-driven models had several drawbacks, including taking a long time to predict and calculate water quality, being difficult to understand, and having inconsistent water quality predictions (Bui et al., 2020). To overcome these challenges, researchers explored AI-based non-physical models, which significantly improved the accuracy of water quality analysis and prediction through machine learning and deep learning (Haghiabi et al., 2018; Kouadri et al., 2021; Prasad et al., 2022). These AI-based models have gained popularity in water quality monitoring because they can predict complex phenomena, handle large datasets, and are not impacted by missing data (Tiyasha et al., 2020; Hanoon et al., 2021).

Recent studies have applied various machine learning algorithms to predict water quality in lakes, groundwater, and rivers (Zhu et al., 2022; Yang et al., 2021; Jiang et al., 2021; Nayan et al., 2020; Chou et al., 2018; Kouadri et al., 2021; Nasir et al., 2022; Afzaal et al., 2019). The most widely used ML algorithms include artificial neural networks (ANN) (Gupta et al., 2019; Vijay and Kamaraj, 2021; Trach et al., 2022; Ubah et al., 2021), adaptive neuro-fuzzy inference systems (Tiwari et al., 2018; Sahoo et al., 2021), random forest (Tung and Yaseen, 2021; Bui et al., 2020; Sakaa et al., 2022), support vector machine (Leong et al., 2021; Chia et al., 2022; Singh et al., 2011), Radial Basis Function (Abba et al., 2020; Leong et al., 2021), Gradient Boosting Machine (Nayan et al., 2020; Khoi et al., 2022), Generalized Linear Model (Dimple et al., 2022; Pujar et al., 2020), among others. For example, Nayan et al. (2020) used the GBM model to predict the WQI of multiple rivers in Bangladesh with high accuracy. Chou et al. (2018) evaluated four AI models, ANN, SVM, CART, and linear regression, to predict the WQI of 20 reservoirs in Taiwan and found that the ANN model was the most effective. Kouadri et al. (2021) applied eight ML algorithms to predict WQI for the Illizi region in southeastern Algeria and found that multi-linear regression outperformed the other models. Yaseen et al. (2018) used several ML models to predict the WQI and found that ML algorithms can solve non-linear problems with high accuracy. However, the performance of these models depends on the input factors, algorithm selection, and output variables, so researchers must evaluate different models under different conditions for various problems.

Machine learning models have been widely used for water quality analysis, but they also have certain weaknesses, such as their high dependence on training and testing data and low prediction accuracy (Bui et al., 2020; Kouadri et al., 2021; Tao et al., 2022). To address these limitations, hybrid models combining machine learning and meta-heuristic algorithms have been proposed (Alqahtani et al., 2022; Song et al., 2021). These hybrid models are more flexible and robust when dealing with noisy data, but they are also complex methods that take a lot of time and require a difficult search for a good meta-heuristic algorithm (Ibrahim et al., 2022).

To overcome these limitations, deep learning models have been proposed for water quality modeling. They can adjust model components frequently and adaptively to deal with highly unpredictable and non-linear predictions (Ewuzie et al., 2022; Prasad et al., 2022). Additionally, they only analyze the relationship between inputs and outputs without considering the causal relationship between variables, making the modeling process more convenient for environmental managers (Wan et al., 2022). However, the stacking ensemble model outperformed other ML and hybrid models under less data condition as it consists of two or more ML models that serve as a base and one ML model as a meta-learner (Madni et al., 2023; Li et al., 2022a; Wang et al., 2021).

Deep learning models, such as DNN and CNN, have been widely used to predict WQI and have been shown to outperform classical ML models, ensemble ML, and hybrid ML models (Nasir et al., 2022; Singha et al., 2021; Barzegar et al., 2020; Prasad et al., 2022; Tung and Yaseen, 2021; Palkar et al., 2022; Uddin et al., 2022; Raheja et al., 2022; Dong et al., 2019; Sheikh Khozani et al., 2022; Kogekar et al., 2021; Jiang et al., 2021). This is because deep learning algorithms have a higher ability to detect non-linear complex relationships (Nasir et al., 2022). Previous studies have shown that the performance of the CNN model can be enhanced by increasing the number of inputs and optimized DL models can perform well with less data (Afzaal et al., 2019; Mallick et al., 2022; Talukdar et al., 2020).

In this research, we introduce a novel approach for predicting Lake WQI by combining machine learning and deep learning models into a stacking ensemble framework. This study also includes the use of optimized DNN and CNN deep learning models, which are tested for their performance when dealing with limited data.

Previous studies have utilized sensitivity analysis to predict water quality, climate, hydrology, and other related factors using methods such as Sobol Indices, the extended Fourier amplitude sensitivity test (eFAST), the Morris one-at-a-time (MOAT) method, and linear regression models (Hsieh et al., 2018; Xue et al., 2021; Li et al., 2022b; Qian and Mahdi, 2020; Liu and Ding, 2020; Dela et al., 2022). These models are effective in determining the impact of input parameters on output parameters.

This study fills this gap by proposing a stacking ensemble framework that predicts WQI, while also conducting sensitivity and uncertainty analysis of parameters affecting WQI using DNN. This represents a new concept and provides a more robust and accurate prediction of water quality compared to traditional methods.

This study has important implications for the development and adoption of cleaner production technologies for sustainable water pollution management. By using the proposed stacking ensemble of machine learning algorithms and optimized deep learning models, it is possible to accurately predict the WQI, which can aid in the development of effective water pollution management strategies. Additionally, the sensitivity and uncertainty analysis of the parameters affecting the WQI can provide valuable insights into the key factors that contribute to water pollution, and this information can be used to guide the implementation of cleaner production technologies that are better suited to the specific conditions of the study area. This approach represents a significant departure from the traditional way of water pollution management, as it provides a more comprehensive and data-driven approach to decision making. Therefore, sustainable management plans may be formulated and implemented more effectively, leading to improved water quality for both human and aquatic ecosystems.

2. Materials and method

2.1. Study area

Lake Loktak, India's largest freshwater lake, is located in the state of Manipur (Fig. 1). The lake is famous for its floating islands, locally

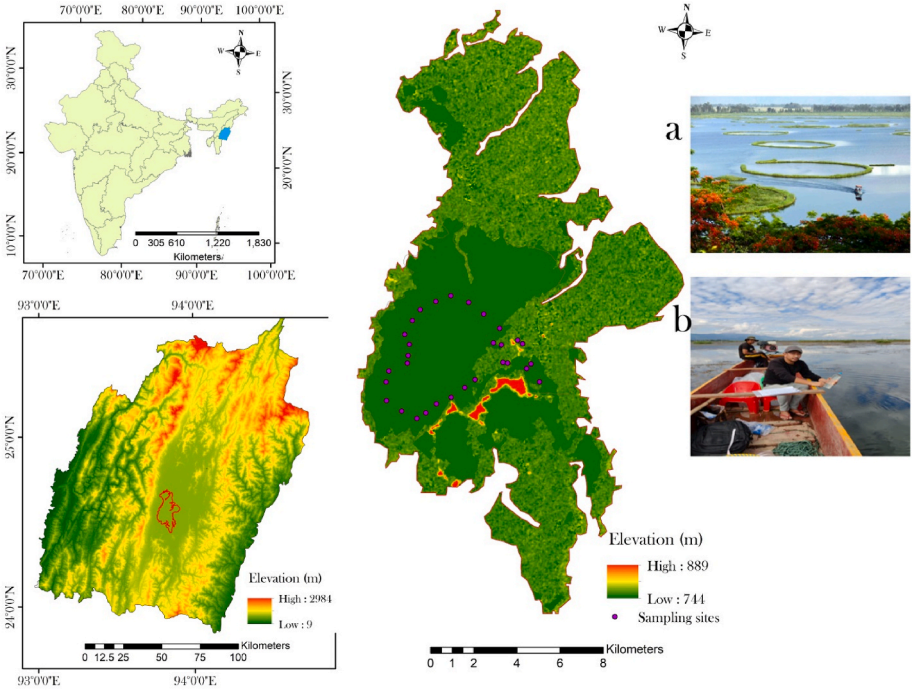


Fig. 1. Location of the study area with sampling points, (a) Phumdis, and (b) sample collection.

known as phumdis (floating biomass). These phumdis cover about 50% of the lake and control many ecological processes of the lake. The catchment area of the lake is over 5040 km^2, while the surface water cover is about 287 km^2. The area varies from 744 m above sea level (a.s.l) in the valley to 2559 m a.s.l. in the hills. The soil of the catchment consists mainly of clay and silt. A field investigation revealed that the topsoil is moderately to slightly acidic (pH ranges from 4.5 to 5.2).

From a hydrological perspective, Loktak Lake depends on nine main rivers: Khuga, Thongjaorok, AwangKhujairok, Nambol, Nambul, Imphal, Kongba, Iril and Thoubal, with the valley floor covered by alluvium deposited in areas of low gradient. The depth of the lake varies from 0.5 to 4.58 m with a mean of 2.7 m. Fig. 1 shows the sampling sites of Loktak Lake. Tremendous urbanisation throughout the catchment area, especially in the Manipur/Imphal Valley region, has led to deterioration of the water quality of the lake in the last few decades. The lake has been formed due to tectonics and neo-tectonics. The average annual rainfall in Manipur varies between 1500 and 1700 mm. The lake area has a humid to subtropical monsoon climate, while the middle valley and higher elevations are semi-tropical and temperate. Temperatures are 24 °C in summer (May–July) and 14 °C in winter (December–February) (November–January). There are no major manufacturing or mining operations in or near the catchment area. A large amount of runoff and domestic waste from the catchment is discharged into Lake Loktak during the rainy season from the approximately 36 rivers and streams that feed the lake. The western catchment provides 52% of the lake's water, while direct rainfall and other inflows provide 25% each, and the remaining 23% comes from the lake itself.

2.2. Sample collection and rationale for selecting the water quality parameters

All sampling sites in this study were selected so as to capture the entire lake characteristics. Water samples were collected from 30 different sites in Loktak Lake in the post-monsoon season (Fig. 1), as indicated in the guidelines from Bureau of *Indian Standards (*BIS). Samples were collected from the Loktak lake, with random selection of sampling sites so as to cover almost entire area. However, it should be noted that at some location, sampling is not possible due to the presence of the phumdis and the fishing zone. A figure of phumdis is shown in Fig. 1a for reference. Also, the samples were taken by boat. In some places the depth of the lake was shallower, so it was not possible to navigate these areas by boat. Therefore, we were able to do sampling at 30 locations. The sampling was done in the post-monsoon season. During post-monsoon, the depth of the lake decreases due to evaporation and lack of rainfall. The Central Pollution Control Board (CPCB) and BIS[1] (10500:2012) guidelines state that the depth of sampling for surface water should be less than 30 cm. However, in this study, the depth of sampling was kept at 20–30 cm. This was because during sampling it was observed that aquatic plants (hydrilla, water hyacinth, water chase nut, water lily, etc.) were growing in the lake at a depth of less than 30 cm (approximately). A portable multiple water analyser (model HI98129, Hanna India) is used to measure the main physical parameters i.e. pH, EC, temperature, turbidity and total dissolved solids (TDS) immediately. The collected water samples are stored in the pre-treated polyethylene bottles at 4 °C and brought to the laboratory for analysis. Prior to analysis, the samples were filtered with 0.45-μm philtres and the American Public Health Association (APHA) standard technique was used to analyze the samples for parameters such as dissolved oxygen (DO), biochemical oxygen demand (BOD), chemical oxygen demand (COD) and nitrate (NO_3^-). We used several methods, such as Winkler acid modification titrimetric, bottle incubation, open reflux and Ultraviolet–visible (UV–Vis) spectrophotometry for the analysis of DO, BOD, COD and NO_3^- respectively. In addition, the average values of the physiochemical properties were compared with the WHO drinking water standards.

2.3. WQI estimation

The WQI has been calculated to investigate the impact of natural and anthropogenic activities on several key parameters of surface water

[1] Central Pollution Control Board, "Guidelines for water quality monitoting."(https://cpcb.nic.in/openpdffile.php?id=UmVwb3J0RmlsZXMvTmV3SXRlbV8xMTZfR3VpZGVsaW5lc29mIHdhdGVycXVhbGl0eW1vbml0b3JpbmdfMzEuMDcuMDgucGRm).

chemistry. Several researchers have conducted surface water quality studies to investigate the impact of anthropogenic activities on surface waters (Noori et al., 2019; Gao et al., 2020; Rahman et al., 2020; Adimalla et al., 2018; Mukate et al., 2019; Kachroud et al., 2019). Following previous research, this study estimated the WQI by integrating several physiochemical parameters. Based on previous literature review, we selected nine parameters (pH, T, EC, TDS, turbidity, DO, BOD, COD and nitrate) based on their importance to surface water quality. The World Health Organisation Drinking Water Standard (WHO, 2011) and the Indian Standard Specification (BIS, 2012) were used to calculate the WQI. The following steps were adopted for the calculation:

The relative weight (RW) for all physiochemical parameters was calculated by using the assigned weights (AW) of all parameters. The assigned weights were calculated based on literature research and expert opinions. The assigned value for nine parameters varies from 1.46 to 4.09, depending on the importance of the parameters for water quality (Table 1). The parameters with critical health impacts were weighted most heavily, as their presence above desirable limits could make the resource less suitable for domestic and drinking water use. Equation (1) was used to calculate the relative weighting:

$$RW = \frac{AW}{\sum_{i=1}^{n} AW_i} \tag{1}$$

Where RW = relative weight, AW = assigned weight of each parameter, and n = number of parameters.

To calculate the WQI, another parameter is required, namely a quality rating scale (Qi) for all parameters. The Qi was determined by dividing the amount of a particular element in the water sample by the standard concentration of the particulate elements according to the standard set by WHO. Moreover, for some parameters, the drinking water standard of WHO is not available, so we used the Indian standard (BIS, 2012). However, the Qi value was calculated using equation (2):

$$Qi = \left(\frac{Ci}{Si}\right) \tag{2}$$

while the quality rating for pH and DO was calculated using equation (3):

$$QipH, DO = \left[\frac{C_i - V_i}{S_i - V_i}\right] \times 100 \tag{3}$$

where, Qi = the quality rating, a Ci = value of the water quality parameters obtained from the analysis, Si = value of the water quality parameter obtained from WHO and BIS parameters, Vi = ideal value for pH = 7.0, and DO = 14.6.

In the next step, the sub-indices are required to compute the WQI using equation (4):

$$SI_i = RQ \times Qi \tag{4}$$

After computing sub-indices values, the final WQI has been computed using equation (5):

$$WQI = \sum SI_i \tag{5}$$

Table 1
Relative weight of the each parameter.

Parameters	Units	WHO standards (2008)	Assign weights	Relative weights
pH	–	6.5–8.5 (7.5)	2.7	0.105
TDS	mg/l	450	3	0.117
EC	ls/m	250	3.5	0.136
Temperature	°C	24	3	0.117
Turbidity	NTU	5	2.5	0.097
DO	mg/l	6	4	0.155
BOD	mg/l	3	4	0.117
COD	mg/l	15	3	0.077
Nitrate	mg/l	5	5	0.077

2.4. Prediction of WQI using ensemble machine learning and deep learning

Here we proposed a stacking model combining three hypertuned algorithms such as GBM, GLM and NN, and used a hypertuned DNN and CNN model to predict the WQI. The modelling phase was initiated by randomly dividing the entire dataset into training and a testing dataset in a 70%–30% ratio. The training dataset (70%) was used for model building and the test dataset was used to validate and test the performance of the models created. The details of the models can be found below.

2.4.1. Stacking ensemble machine learning algorithm

In this study, we have proposed a stacking framework to predict the WQI of Lake Loktak. The stacking framework is a kind of robust ensemble of machine learning algorithms that works in two ways to combine multiple machine learning algorithms (Wolpert, 1992). The stacking ensemble model has been successfully applied in various research areas (Zounemat-Kermani et al., 2021; Kshatri et al., 2021; Lasisi and Attoh-Okine, 2019; Di et al., 2019; Cui et al., 2021). First, multiple machine learning algorithms (ML) are implemented to predict the target variables, which is referred to as base-learner (level-0). Then, the prediction of multiple MLs is integrated by another ML algorithm, which is called meta-learner (level-1). In this study, we predicted the WQI using Gradient Boosting Machine (GBM), Generalized Linear Model (GLM) and Neural Network (NN) as base-learner (Level-0) and combined all algorithms with GLM as meta-learner (default meta-learner in h2o framework) (Level −1). The theoretical background of the meta-learner and base-learners is explained below.

2.4.1.1. Gradient Boosting Machine (GBM). The GBM model is a tree-based ensemble model that combines multiple decision trees through gradient-based optimization and boosting algorithms (Liang et al., 2020; Sahin, 2022; Adi et al., 2020), which improves the prediction performance (Rong et al., 2020; Di et al., 2019). The datasets were divided into different subgroups by a single decision tree model (Chen et al., 2019). However, in the case of the GBM, a sequence of multiple decision trees was iteratively constructed by the GBM to minimise loss (error between actual and predicted data) (Huang et al., 2019). The details of the mathematical basis and resampling procedure of GBM can be found in the work of Friedman (2001, 2002). Many previous researches have shown that GBM has achieved higher accuracy in predicting WQI compared to other ML algorithms (Khan et al., 2022; Lu and Ma, 2020; Ahmed et al., 2019; Nasir et al., 2022).

2.4.1.2. Generalized Linear Model (GLM). The GLM model is an extended concept of the widely used linear regression model. It provides an overall function that can handle linear and non-linear data sets for regression analysis (Pataca et al., 2020; Ghorbani et al., 2020). It generates the model from the exponential family for output. The estimation of the model can be achieved through the parameter vector of observations by maximising the log-likelihood (Ghorbani et al., 2020). Although it is a simple model, it has been widely used to predict WQI in previous research (Mohammed et al., 2020; Kouadri et al., 2021; Araromi et al., 2018; Elbeltagi et al., 2022). Therefore, we used GLM for predicting WQI in the present study.

2.4.1.3. Neural network (NN). Recently, NN algorithm has gained much popularity over machine learning algorithms due to its higher accuracy, ability to process large and nonlinear datasets, training stability, ability to solve complex nonlinear problems and hierarchical feature selection (Tung and Yaseen, 2021; Singha et al., 2021; El Bilali et al., 2022;

Costache et al., 2022). In this study, a multi-layer feed-forward deep learning architecture was used in the framework of h2o (Ghorbani et al., 2020). This algorithm uses architecture similar to the conventional multilayer perceptron (MLP). The weights of the input parameters were calculated in each neuron and passed to the next neuron using a transfer function (Ghorbani et al., 2020). This model consists of three layers, namely the input layer, several hidden layers and the output layer. The h2o framework is a sophisticated artificial neural network with back-propagation optimization (stochastic gradient decent). In this research, the deep learning algorithm was tuned to the training data in the h2o framework using the random grid search technique with 5-fold cross validation.

Finally, all three machine learning models were combined in the stacking framework. In this study, we again used the GLM algorithm as the meta-learner. Therefore, all base-learners were integrated using GLM -based meta-learners to predict the WQI.

2.4.2. Deep neural network (DNN)

In this research, we implemented the DNN algorithm using the Keras package in the R programming language (Choi et al., 2021; Zou et al., 2020; Singha et al., 2021). However, the theoretical background of DNN is almost similar to that of NN, which is explained in Section 2.4.1. However, the difference is the use of number of hidden layers. Moreover, the DNN algorithm has a considerable number of hidden layers that can predict highly nonlinear complex data sets. In this study, we used a Rectified Linear Unit (ReLU) activation function for the hidden layers and a linear activation function for the output layers. The hyper-parameters of the architecture were adjusted using the cost function and the optimization algorithm until the actual observations matched the predicted value.

2.4.3. Convolutional neural network (CNN)

A CNN is a type of neural network that extracts patterns from data by applying convolutions, with the aim of discovering filters that obtain the most relevant features from the input to perform a specific task (Yan et al., 2021; Liu et al., 2022; Di, 2022; Pang et al., 2020). Therefore, this algorithm is able to automatically detect new elements and resize them by learning from the data pattern (C. Zhang et al., 2022). As a result, researchers have referred to this algorithm as 'object-recognising organisms' that functions for object detection and image classification (Di, 2022). However, the architecture of the CNN algorithm is quite different from other neural networks as it consists of a convolutional layer and a pooling layer (Song and Kim, 2020). These layers can effectively reduce the complexity of the original network and the large number of parameters, which enables more efficient forward propagation. The local time feature of the data can be detected by the convolution kernel of the convolution layer through sliding the window, and the weights can be obtained from the sliding convolution kernel (Song and Kim, 2020). Furthermore, the pooling layer is used for dimensionality reduction by abstracting the different feature extractions through multiple convolution kernels. Finally, the global feature was generated by integrating local features using fully connected layers. Normally, the CNN model is used for spatial and geometric feature extraction with two- and three-dimensional input data, while it also works perfectly with one-dimensional data. Therefore, to run this one-dimensional CNN model, the one-dimensional convolutional layer should be used (Di, 2022). Previous researchers have successfully used this 1D CNN model and achieved higher accuracy (Baek et al., 2020; Pyo et al., 2020; Sha et al., 2021; Choi et al., 2021; Khan et al., 2022; F. Zhang et al., 2022). In this study, we used a 1D CNN model because our data structure is one-dimensional.

2.5. Evaluation of the models

The performance of the models was evaluated using several functions, such as the percentage of predictions that were within a factor of two of the true concentrations (FAC2), the normalised mean bias (NMB) and the normalised mean gross error (NMGE). The root mean square error (RMSE), the correlation coefficient (R), fractional bias (FB), the coefficient of efficiency (COE) and the index of agreement (IOA) are some of the other evaluation methods.

The RMSE measures how well the simulated values match the observed values. An over or underestimating prediction is often represented by the mean bias (MB). To calculate the NMB, we divide the value MB by the measured concentration. While NMB takes into account whether a prediction is too high or too low, NMGE does not (absolute value). The correlation coefficient measures the linear relationship between two variables. To determine whether the model has a-systematic tendency to over or under-predict, FB is used. The optimal value of FB is zero, and its range is between -2 and $+2$. If the value is negative, the model overestimates the forecasts, and if it is positive, it underestimates them. FAC2 measures how often the predicted values are within two standard deviations of the observed values. In other words, FAC2 is a count of the proportion of points that are within 0.5 and 2 times the observed values and meet the 0.5 Mi/Oi 2.0, where Mi and Oi are the modelled and observed WQI values. See Carslaw (2011) and Derwent et al. (2010) for more information on the mathematical formulations behind these measures.

2.6. Sensitivity and uncertainty analysis using a deep neural network-based sensitivity method

In this study, to the best of the author's knowledge, this is the first time we have used a DNN-based sensitivity and uncertainty analysis for WQI prediction. Many previous research papers have used sensitivity analysis for forecasting water quality, climate, hydrological variables and other areas. They used variance-based sobol indices, extended FAST, Moris OAT, linear regression-based models (Hsieh et al., 2018; Xue et al., 2021; Li et al., 2022b; Qian and Mahdi, 2020; Liu and Ding, 2020; Dela et al., 2022). These models are also very effective in evaluating the impact of input parameters on the output parameter. But neural networks and DNN based on sensitivity and uncertainty analysis have not yet been used in water resources studies.

In this study, we implemented this model in two ways. First, we implemented a DNN and imported the model into a sensitivity analysis system. The theoretical background of DNN was explained in the previous section on methodology (section 2.4.2). After determining the best DNN model, the weights and the neural structure of the model were extracted using function 'get_weights ()' of the package 'keras; '. Then we used function 'SensAnalysisMLP ()' of the package 'Neuralsens' to perform the sensitivity and uncertainty analysis for the extracted object and the same datasets were trained with DNN. In this way, three sensitivity measures of the output (predicted WQI by the stack algorithm) were calculated with respect to the input parameters. The sensitivity measures are (a) the mean effect of the input variables on the output, (b) the variance of the effect of the input variables on the input, and (c) the measures of the input parameters from a perturbation analysis point of view, which means that larger changes in the output may occur when small changes were made to the input parameters. However, the meaningful information can be derived from the relationship between the mean and variance effect of the input parameters on the output parameters. The relationship may be as follows:

If both the mean and variance of the input parameters are towards zero, there is no relationship between input and output because the sensitivity of the output to that particular input goes to zero.

If the mean of the input has a value other than zero and the variance goes towards zero, it can be called linear relationship between input and output, since the sensitivity of the output with respect to that particular input is approximately constant.

The uncertainty analysis was calculated using the density plot considering the mean and standard deviation of the sensitive value of the output, which shows the uncertainty of the output with respect to the

data distribution pattern of the input parameters. The coefficient of variation (CV) was calculated to analyze the degree of variance of the data distribution pattern for the uncertainty analysis.

3. Results

3.1. Descriptive analysis of water quality parameters

In this study, nine physiochemical parameters of the surface water of Loktak Lake, Manipur are used to calculate and predict the WQI. Laboratory analysis of physico-chemical properties reveals considerable insights of surface water quality. In this study, we have identified the parameters that are most responsible for the deterioration of water quality using the basic statistical analysis. This study revealed an average pH of 8.61, with a minimum of 7.7 and a maximum of 10, indicating the alkalinity of the water bodies (Table 2). The average TDS value in this study is 68.6nullmg/L, with a minimum value of 45nullmg/L and a maximum value of 86nullmg/L, which is far below the norms of WHO. Therefore, the TDS value in this study is within the acceptable limits. As per WHO standard the average values of EC, turbidity and BOD in this study are within the acceptable limit. On the other hand, the maximum value of turbidity (5.66 NTU) and BOD (5.6nullmg/L) was measured outside the WHO drinking water standards (Table 2). Therefore, it can be concluded that only a few samples have an acceptable limit for turbidity and BOD causes a lower average value, indicating polluted water quality. Furthermore, average values for temperature, DO, COD and nitrate were found to be above the upper limit of the WHO drinking water standards (Table 2). The maximum values of these parameters are far from the WHO drinking water standards, indicating contamination of surface water by pollutants.

3.2. Computation of WQI and its relationship with associated parameters

In this study, the WQI was computed for each sample using an arithmetic weighted approach based on the relative weights (Table 1) of the nine variables. The results showed that DO had the highest weight, followed by EC, pH, and turbidity. Meanwhile, COD and nitrate had the lowest weights. The calculated WQI indicated that the Loktak Lake had a minimum WQI value of 90.75 and a maximum of 145.29, with an average of 112.62 (Fig. 2). According to Yadav et al. (2010), the WQI values can be classified into five levels: excellent (0–25), good (26–50), poor (51–75), very poor (76–100), and unsuitable (above 100). When comparing the results of this study with the classification scheme of Yadav et al. (2010), it was found that most of the samples were classified as very poor and some as unsuitable. 70% of the sample sites were considered unsuitable for water quality, while 30% were considered very poor. This is a concern as Loktak Lake, a designated Ramsar site, is crucial for both biodiversity and humans, but its surface water quality is unsuitable, posing a threat to aquatic ecology and requiring immediate monitoring and management.

Additionally, we conducted a correlation analysis between the water quality parameters and the WQI (Fig. 3). The results reveal a negative correlation between the WQI and EC, TDS (moderate), nitrate, and COD, indicating that higher values of these parameters result in lower WQI values. Conversely, a positive correlation between the WQI and pH (strong, at a 0.05 level of significance), DO (strong, at a 0.05 level of significance), BOD (strong, at a 0.05 level of significance), turbidity, and temperature was found, meaning that higher values result in a higher WQI. Therefore, the management plans can be formulated based on the relationship between WQI and parameters.

3.3. Stacking model implementation

In this study, we implemented a stacking framework that combined GBM, GLM, and NN models. We implemented this framework in two ways: first, by implementing GBM, GLM, and NN with hyper-tuned parameters, and second, by integrating all three models into the stacking framework using the GLM meta-learner. The entire implementation process was done in the R programming language using the 'h2o' package.

The GBM model was hyper-tuned using the random grid search algorithm from the 'h2o' package. We obtained the best parameters for GBM as follows: Number of trees: 54, number of internal trees: 54, minimum depth: 4, maximum depth: 9, mean depth: 5.425, minimum leaves: 6, max leaves: 17, and mean leaves: 10.98. The error matrices (MSE: 85.66, RMSE: 9.25; MAE: 6.61; RMSLE: 0.073) also indicated that the model performed well with these best parameters. Fig. 4a shows the weighting of all input parameters calculated by the model, with pH being the most important for WQI, followed by TDS, EC, and turbidity, while nitrate had the least importance.

After training the GBM model, we implemented the GLM model using the 'h2o' package. This model was also hyper-tuned using the random grid search technique. We obtained the best parameters for the GLM model as follows: family: Gaussian, link: identity ridge, regularization: lambda (0.1335), lambda search: nlambda- 30, lambda.max-1334.7, lambda.min- 0.1335, lambda.1se - 0.1834, and number of iterations- 30. The coefficients for all parameters with this model are as follows: Intercept: 60.45, pH: 16.35, TDS: 0.14, EC: 0.042, temperature: 0.34, turbidity: 1.35, DO: 0.81, BOD: 1.02, COD: 0.24, and nitrate: 2.72.

Based on the above parameters and model structure, the WQI predicted by GLM was evaluated using error matrices, which showed good results (MSE: 71.16, RMSE: 8.43, MAE: 6.02, RMSLE: 0.06). Additionally, the importance of all parameters was calculated based on the model structure (Fig. 4b). It showed that pH had the highest importance, followed by turbidity and nitrate, while EC and TDS had the least importance. The importance also indicates the story of the coefficient of this model. In the same way, we tuned the model NN with the package 'h2o'. The best optimal parameters for NN are:

Layer 1 has 9 neurons, Layer 2 has 20 neurons, and Layer 3 (the output layer) has 1 neuron. The dropout rate for Layer 1 (input) is 20%, and the rectifier dropout rate for Layer 2 is 50%. The dropout rate for Layer 3 is linear (NA). The mean weight for Layer 2 and 3 is −0.35 and −0.019, respectively. The weight_rms for Layer 2 and 3 is 0.317 and 0.194, respectively. The mean_bias is 0.183 and 0.149, and the bias_rms for Layer 2 and 3 is 0.184 and 0.00.

The error matrices for evaluating the predicted WQI by the NN, based on the training data, show satisfactory performance (MSE: 33.74, RMSE: 5.8, MAE: 4.02, RMSLE: 0.04). Based on this model structure, the

Table 2
Descriptive statistics of the water quality parameters and computed WQI.

Descriptive statistics	pH	TDS	EC	Temperature	Turbidity	DO	BOD	COD	Nitrate	WQI
Minimum	7.70	45.00	86.00	9.40	0.02	3.60	1.60	23.00	0.57	90.75
Maximum	10.00	86.00	171.00	30.00	5.66	9.80	5.60	42.00	7.46	145.29
Range	2.30	41.00	85.00	20.60	5.64	6.20	4.00	19.00	6.90	54.54
Sum	258.30	2059.00	4038.00	791.40	85.50	190.8	121.40	935.00	60.43	3378.71
Median	8.45	68.50	132.50	26.85	2.68	6.45	4.30	32.00	1.60	110.04
Mean	8.61	68.63	134.60	26.38	2.85	6.36	4.05	31.17	2.01	112.62
Standard deviation	0.71	8.35	17.25	3.58	1.92	1.53	1.12	4.36	1.33	15.61
Coefficient of variation	0.08	0.12	0.13	0.14	0.67	0.24	0.28	0.14	0.66	0.14

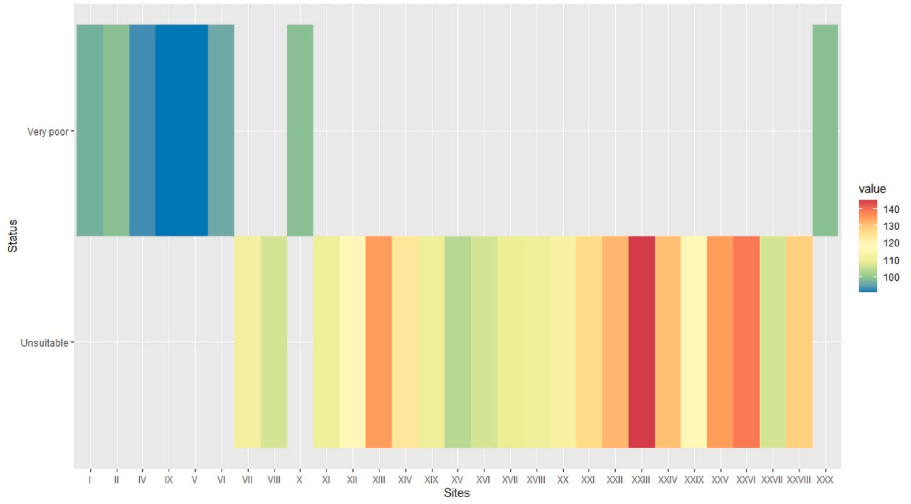


Fig. 2. Computation of WQI for each sample with the pollution status in the Loktak Lake.

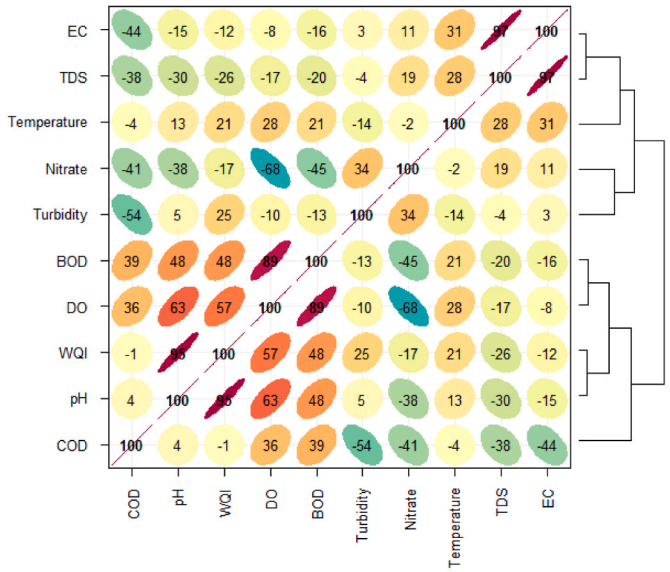


Fig. 3. Correlation coefficient analysis among the water quality parameters and calculated WQI.

importance of all parameters was calculated using the NN and is shown in Fig. 4c. The figure indicates that pH has the highest importance, followed by turbidity and temperature, while COD has the lowest importance, followed by TDS.

After obtaining three models with hyper-tuned parameters, we combined them using the best GLM models (meta-learners) in the stacking context. We evaluated the performance of the model using error matrices and found that the Stacking model has a lower MSE value of 1.29, with an RMSE of 1.13, MAE of 0.96, RMSLE of 0.01, and a mean residual error of 1.29. These results demonstrate that the Stacking algorithm has lower error than any individual models (GBM, GLM, and NN). This combination of stacking algorithm with machine learning and deep learning algorithms provides a powerful solution.

3.4. DNN implementation

In this study, we trained a DNN model to achieve high accuracy in predicting the WQI. To do so, we implemented the DNN model in three steps. Firstly, we implemented the DNN model with default functions using the 'h2o' package. Then, we created two hidden layers with 9 neurons, which correspond to the number of parameters. Finally, we used the random grid search technique to find the optimal parameters of the DNN. Based on the optimal parameters, we implemented the final model to predict the WQI with high accuracy. However, in the case of the default DNN model, we used two hidden layers with a rectifier activation function and one output layer with a linear activation function. We did not use any dropout or regularization techniques. Based on this model structure, we obtained an MSE of 27.31, RMSE of 5.22, and MAE of 4.26. We also calculated the importance of the variables using this architecture (Supplementary Fig. 2a). The figure shows that pH has the highest importance, while the other parameters have about the same importance. Therefore, we cannot claim that this result can provide accurate predictions.

To improve the model, we changed the number of neurons in the hidden layers to 10 and 10, which provided relatively good accuracy on the training data, such as MSE of 14.42, RMSE of 3.79, and MAE of 3.04. Furthermore, this improved model also changed the importance of the variables (Supplementary Fig. 2b). It showed that BOD had the highest importance of 0.9, followed by COD (0.6), DO (0.58), and pH (0.57). On the other hand, temperature and nitrate had the least importance in predicting the WQI. This result differs significantly from the standalone models. Therefore, we implemented the DNN model again using the random grid search technique.

To obtain a well-tuned DNN model, we hyper-tuned several parameters of the DNN architecture, including the activation function ("Rectifier", "maxout", "maxoutWithdropout", "Tanh", "RectifierWithDropout", "TanhWithDropout"), number of hidden layers with neurons ((5, 5, 5, 5, 5), (10, 10, 10, 10), (50, 50, 50), (100, 100, 100)), epochs (20, 50, 100, 200), regularization_l1 (0, 0.00001, 0.0001), regularization_l2 (0, 0.00001, 0.0001), rate (0.01, 0.005, 0.001), rho (0.9, 0.95, 0.99, 0.999), epsilon (1e-10, 1e-8, 1e-6, 1e-4), momentum start (0, 0.5), and momentum stable (0.99, 0.5, 0). We then performed a random grid search with these combinations, generating 373 models (Fig. 5). All models were sorted by MSE in ascending order. We then plotted the top ten models, along with the last two models (corresponding to the worst performing), based on the MSE during the training period and iterations. The plot shows that the top models with fewer iteration achieved a lower MSE, while the worst models had a very high MSE, even with a higher number of iterations.

However, the best model had the following structure: five layers (one input, three hidden, and one output layer), each hidden layer had 100 neurons with a tanh activation function, and the regularization_l1 and l2 for the three hidden layers and one output layer had values of 0.000100 and 0.000010. Based on this model structure, we obtained an MSE of 6.03, an RMSE of 2.45, and an MAE of 1.46. Therefore, it can be said that

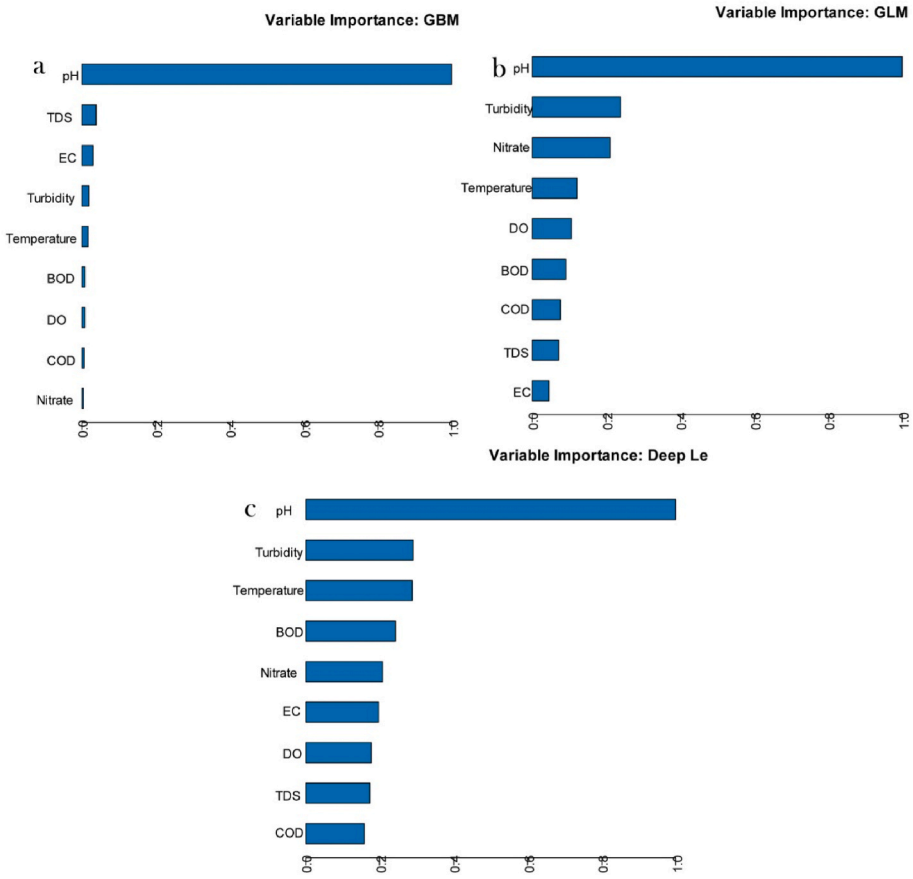


Fig. 4. Computation of importance for water quality parameters using (a) GBM, (b) GLM, and (c) NN.

this model provides quite good results.

3.5. CNN implementation

In this study, we implemented a 1D_CNN model using the "keras" package in the R programming language. Initially, we implemented a simple CNN with two hidden layers of 16 neurons each. We used ReLU activation for both layers, while linear activation was used for the output layer. We also used the RMSprop algorithm for optimization. To evaluate the accuracy of the trained model, we used the mean squared error and mean absolute error metrics, and plotted them against each epoch of the training and validation phases (Fig. 6a). The plot shows that the training and validation error stopped at 20 epochs.

Next, we re-implemented the 1D_CNN model with a larger number of neurons in the hidden layers. The plot of the loss function showed that the error stopped at 30 epochs, which indicates that the model needed to be re-implemented (Fig. 6b). To improve the model, we added four hidden layers with 128, 64, 32, and 16 neurons, along with a 1D layer with maxpooling and dropout to the architecture. We used ReLU activation for the hidden layers and linear activation for the output layer, and the RMSprop algorithm for the loss function. The plot of the loss function (Fig. 6c) shows that a loss at 20 epochs was achieved with a lower value in both the training and testing phases. The evaluation metrics also demonstrate good performance of the improved model.

We assessed the variable importance for two different CNN architectures and plotted the results in Fig. 7a and b. For the optimized CNN architecture, pH was found to be the most important variable for predicting WQI, with a score of 0.9. This was followed by temperature with a score of 0.6, and then nitrate and DO. In contrast, the simple CNN architecture showed that BOD, COD, DO, and pH had the highest importance, while temperature and nitrate were of lower importance. Based on these findings, we can conclude that the optimized CNN performs comparably to other algorithms when it comes to assessing variable importance.

The calculation of variable importance using CNN architectures can provide valuable insights into the factors that have the greatest impact on water quality. This information can be used to inform management decisions and develop effective management plans to address water quality issues.

For example, in the case mentioned where pH was found to be the most important variable for WQI prediction, management efforts could focus on addressing factors that affect pH levels in the water. This could include measures such as reducing the discharge of acidic pollutants into the water or implementing treatments to adjust the pH levels.

Similarly, if BOD and COD are found to be important variables for water quality, management strategies could focus on reducing the discharge of organic matter and pollutants that contribute to these variables. This could involve improving wastewater treatment processes, reducing nutrient inputs into water bodies, and implementing measures to reduce erosion and sedimentation that can contribute to organic matter inputs.

3.6. Evaluation of the models

In this study, we predicted the water quality index (WQI) using six different models, namely GBM, GLM, NN, stacking (a combination of the three models mentioned), DNN, and CNN, for both training and testing datasets. The predicted WQI values for each sample in the training and testing datasets were evaluated using various metrics, including correlation coefficient, FAC2, MB, MGE, RMSE, NMB, NMGE, COE, and IOA. Fig. 8a shows the correlation coefficient between the actual WQI and the predicted WQI for the training periods for all models. It indicates that

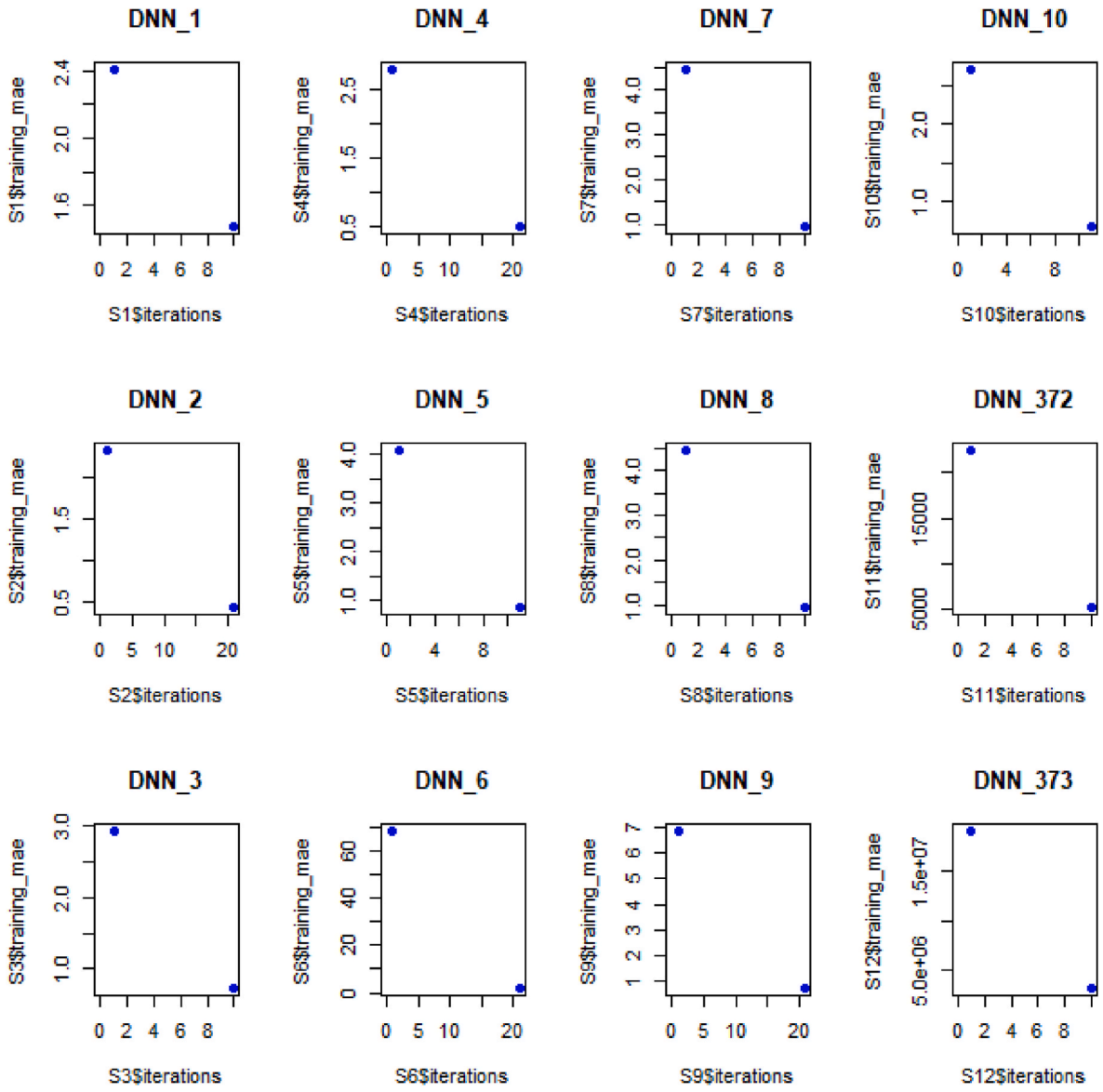


Fig. 5. Evaluation of the performance of DNN models based on MSE during training phases and number of iterations.

the stacking model has the highest correlation coefficient (100%) for both the training and testing datasets between the actual and predicted WQI, followed by CNN (99%), DNN (99%), NN (98%), GBM (98%), and GLM (91%). This result shows that the performance of all six models in predicting WQI based on the training dataset is very satisfactory. Moreover, the comparison between the models shows that the correlation between GLM and NN has the highest value of 93%, followed by CNN (92%), stacking (91%), DNN (90%), and GBM (88%). It can be observed that all the models have a difference in correlation value of 1%–10%, which demonstrates that all the models have good performance.

We also evaluated the WQI predicted by the six models with the actual WQI based on the test dataset using the correlation coefficient (Fig. 8b). It shows that the stacking model has the highest correlation value of 98%, followed by CNN (93%), DNN (83%), GLM (83%), NN (79%), and GBM (75%). This result indicates that the stacking model outperforms all other models, while the difference between stacking and CNN is only 5%. Thus, it can be concluded that stacking and CNN are excellent models for predicting WQI, whereas GBM has only a 75% correlation with actual WQI, which cannot be considered satisfactory. However, this may be caused by under-fitting.

The evaluation of the prediction of WQI by six models based on the training dataset in terms of FAC2, NMB, NMGE, COE, and IOA (Fig. 9) and MB, MGE, and RMSE (Supplementary Fig. 3) shows satisfactory performance. A perfect model achieves a FAC2 value of 1. In the present study, all the models achieved a FAC2 value of 1 based on the training dataset, indicating that all the models perfectly predicted the actual WQI. In the case of the NMB statistic, the expected value is 0, which can be considered a perfect model that neither underestimates nor overestimates. Fig. 9 shows that, with the exception of GBM and GLM, which exceeded 0.05, all models have an NMB value of 0. This indicates that all models predicted the WQI perfectly and did not overestimate or underestimate, while GBM and GLM had a very slight increase in the NMB value, although we cannot consider them as overfitting models. Lower values (<0.2) of NMGE indicate good performance of all models in predicting the WQI. Compared to the other models, NN and GBM have a slightly higher value, indicating that these two models perform very well but not like the other four models. The values of COE and IOA range from -1 to 1, with a value of 1 considered perfect and of high quality for both statistics. The plot of COE shows that the stacking model achieved a value of >0.87, followed by CNN (0.87). Based on the analysis, we can consider these models to be almost perfect and of good quality. The

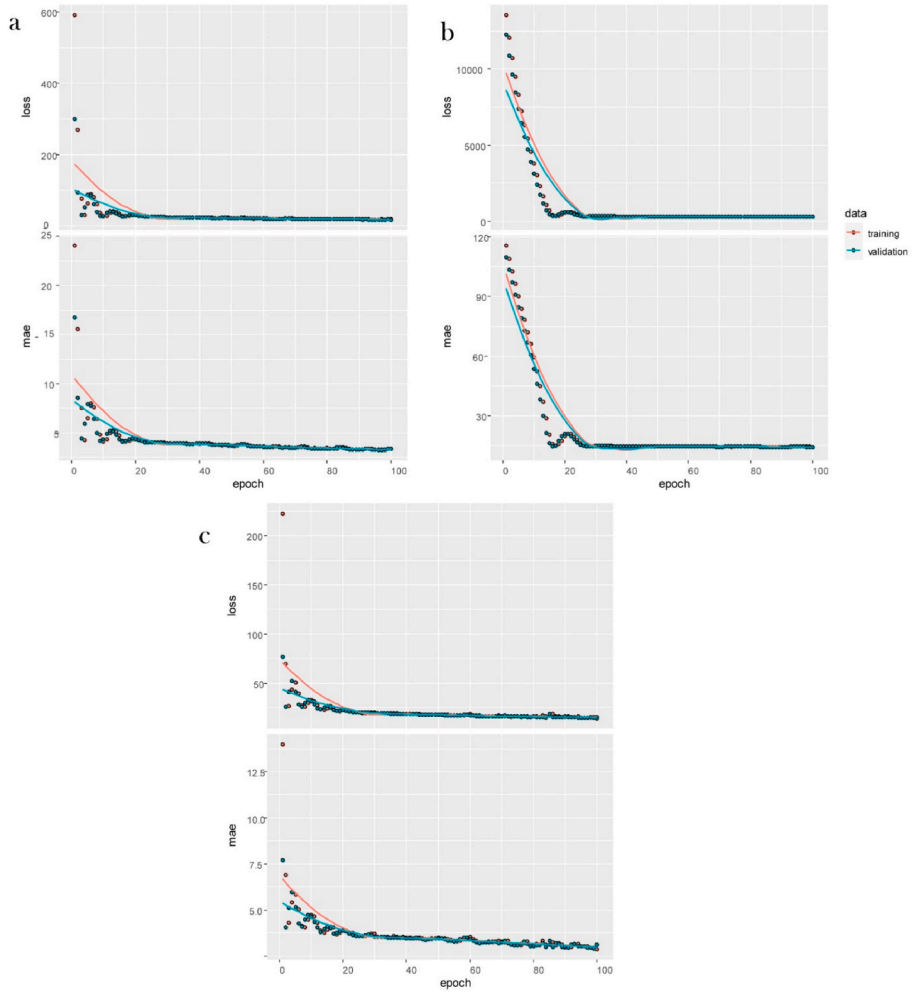


Fig. 6. Training and validation accuracy changes with the number of epochs using CNN with (a, b) simple architecture to (c) hyper-tuned architecture.

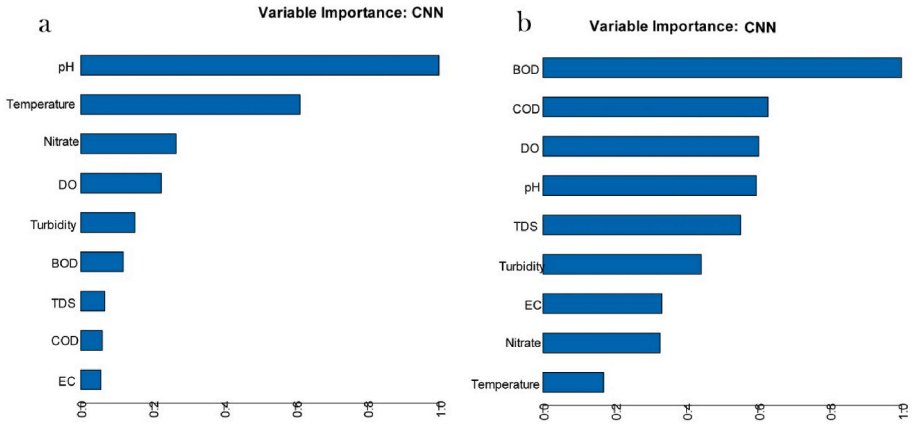


Fig. 7. Computation of importance for all water quality parameters using (a) optimized CNN, and (b) simple CNN.

other four models have a COE value of 0.47–0.52, indicating a medium quality model. Moreover, the IOA value for the stacking model is > 0.88, followed by CNN (0.87), while the IOA value for the other four models ranges from 0.72 to 0.76, indicating that these models are of moderate quality in predicting the WQI.

The stacking model had the lowest RMSE (1.6) followed by CNN (2.4), while the GBM model had the highest RMSE (8.3) followed by NN (7.5) (Supplementary Fig. 3). Thus, the overall analysis of predicting WQI based on the training dataset showed that the stacking and CNN models perform better, while the GBM and NN models perform well but not satisfactorily.

Similarly, we evaluated the performance of six models in predicting WQI based on the test data set using FAC2, NMB, NMGE, COE, and IOA statistics (Fig. 10), as well as MB, MGE, and RMSE (Supplementary Fig. 4). As shown in Fig. 10, all models achieved a FAC2 score of 1, indicating perfect prediction of the actual WQI. Fig. 10 shows that the stacking and

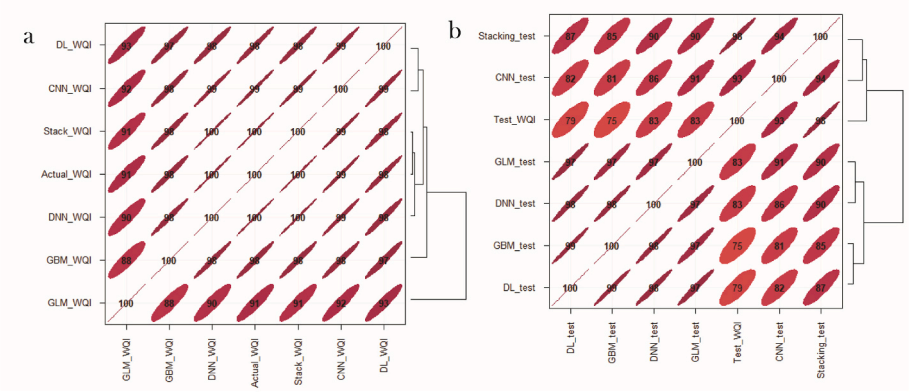


Fig. 8. Correlation coefficient analysis between actual WQI and predicted WQI by GLM, GBM, NNNN, Stacking, DNN, and CNN for (a) training and (b) testing phases.

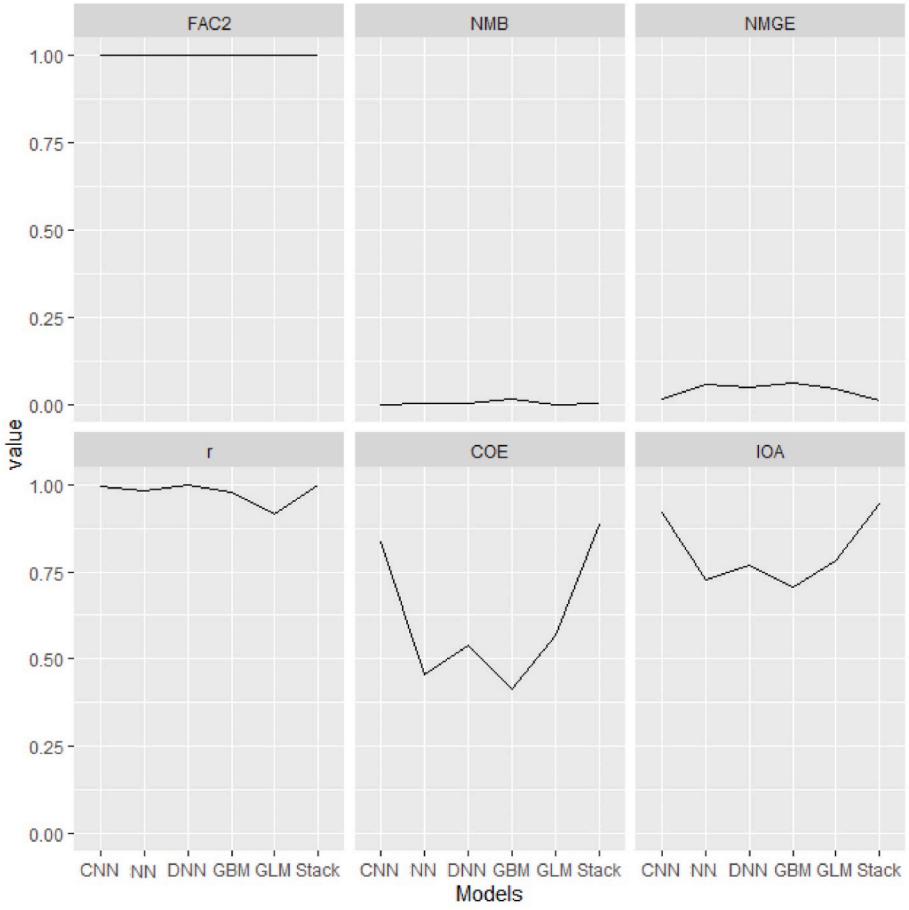


Fig. 9. Evaluation of the models performance for predicting WQI based on training dataset using FAC2, NMB, NMGE, correlation coefficient, COE, and IOA.

CNN models achieved a value of 0, indicating perfect models, while the other models showed slight underfitting in prediction. Moreover, all models demonstrated a NMGE value of less than 0.12, indicating good performance in prediction. In terms of COE, only the stacking model achieved a COE value of 0.75, followed by CNN (0.74), while the other models showed a COE value of less than 0.5, indicating moderate performance, which could be due to insufficient fitting of the prediction. Additionally, IOA statistics showed similar results to COE. The lowest RMSE was found for the stack model (5), followed by CNN (7.5). On the other hand, the maximum RMSE was found for the GBM and GLM models (12.5), followed by NN (11) (Supplementary Fig. 3). Therefore, the overall analysis of WQI prediction based on the test data set demonstrated that the stacking and CNN models performed better, while the GBM, GLM, and NN models exhibited poor prediction fit.

3.7. Comparison of the models

In this study, the performance of the model was compared graphically using the robust Taylor diagram. It graphically explains how well the models match the target variable or WQI. This diagram quantifies the correlations between the actual observation and the predicted observations in terms of correlation coefficients and standard deviations.

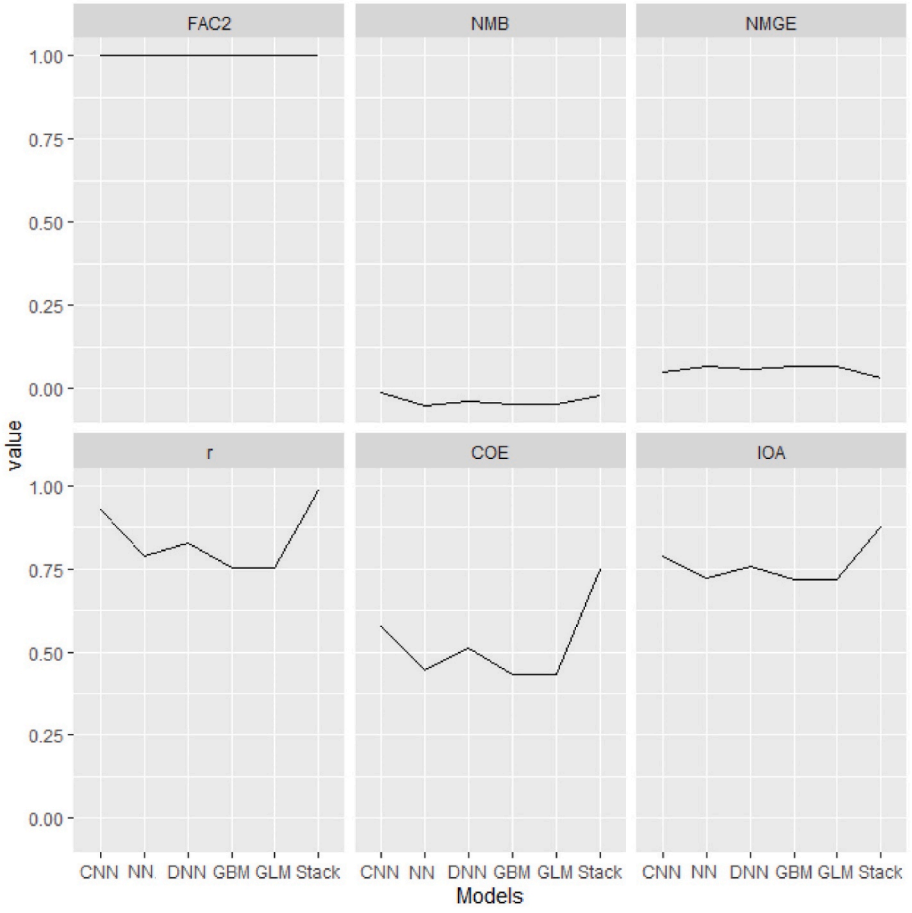


Fig. 10. Assessment of the performance of the applied models based on testing dataset using FAC2, NMB, NMGE, correlation coefficient, COE, and IOA.

In the present study, WQI was estimated using GLM, GBM, NN, Stacking (combination of these three models), DNN and CNN models. We compared all these models for training and testing datasets to investigate the predictive ability of the model for both phases (Fig. 11). The distance to the observed (actual observation) shows the measurement of the centred RMS error. Therefore, we can consider a model as the best model if it is very close to the observation point due to a high correlation value (towards 1), low RMSE and low variance. In the present study, the stacking model outperformed the other models for training phase, including CNN and DNN, in predicting WQI due to high correlation (>0.98) and low RMSE (<2), followed by CNN (correlation: >0.97, RMSE: <3) (Fig. 11a). Based on predictive ability, the models can be ranked in the following order: stacking > CNN > DNN > NN > GBM > GLM. In predicting the test data, the stacking model again proved to be the best model, achieving a high correlation value of >0.97 and a low RMSE of <5, followed by CNN (correlation: >0.9 and RMSE: <8) (Fig. 11b). All models can be ranked based on their predictive power as follows: stacking > CNN > DNN > NN > GLM > GBM. Based on the

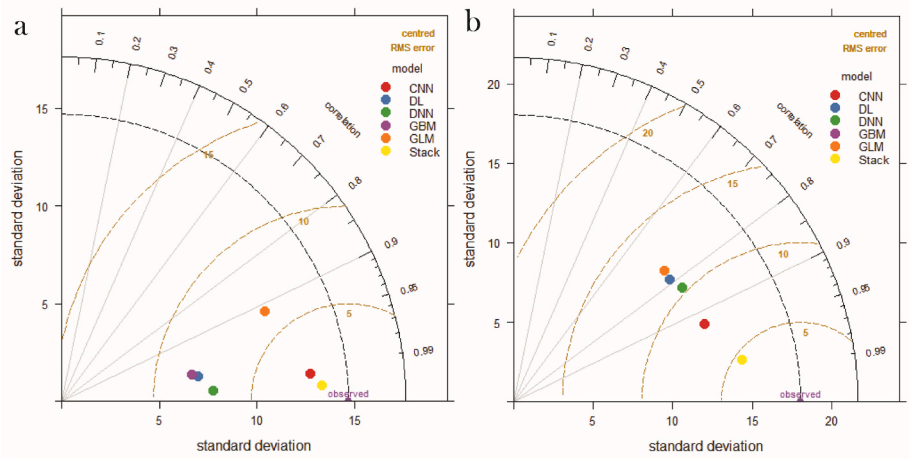


Fig. 11. Taylor diagram for the graphical presentation of the performance of six models, namely, GLM, GBM, NN, DNN, CNN, and stacking for (a) training, and (b) testing dataset.

comparative analysis for both phases, it can be seen that the stacking algorithm outperforms the other models, followed by the CNN model.

3.8. Sensitivity and uncertainty analysis deep neural network sensitivity measure

In this study, a DNN model was used to conduct a sensitivity analysis to assess the sensitivity and uncertainty of the parameters with respect to the output. The process was implemented using the R programming language and various packages such as 'keras', 'tensorflow', 'Neuralsens', and 'monmlp'. The target variable was the WQI predicted by the stacking algorithm, and the nine water quality parameters were the input variables. The stacking algorithm was chosen as the target variable as it outperformed other machine learning and deep learning models. The DNN model was trained using hyper-parameter tuning, which involved finding the best number of neurons (16, 32, 64, 128) and the best dropout rate (0.01, 0.02) using a random search technique. The algorithm was composed of five fully connected layers and dropout layers, and the ReLU activation function was used for the hidden layers, while the linear activation function was used for the output layers. The RMSprop algorithm was used for optimization. The accuracy of the trained model was evaluated using mean square error and mean absolute error matrices and was plotted against each epoch of the training and validation phases. As seen in Fig. 12, the error decreases rapidly in the first two epochs, indicating that the network learns quickly. Afterward, the curves level off, suggesting that further training of the model is not necessary. Hence, the optimal model with 150 epochs was selected.

However, this study includes a label plot, bar chart, and density plot to illustrate the sensitivity and uncertainty of the output with regards to the input parameters (Fig. 13). The label plot displays the correlation between the mean and variance of the impact of the input parameters on the output. It reveals that TDS has a non-linear, negative relationship with WQI (the mean of the sensitivity is less than 0 and the variance of sensitivity is 0.05). On the other hand, the label plot demonstrates that all parameters have non-linear effects on the output, however, the level of non-linearity differs. For instance, pH and turbidity show high non-linearity, followed by COD, temperature, nitrate, EC, and BOD. However, this relationship does not provide any information about the extent of influence or importance of the parameters for the output. The bar chart shows the sensitivity measures of the input parameters in relation to the output parameters. When small adjustments were made to the input parameters, significant changes could be observed in the output.

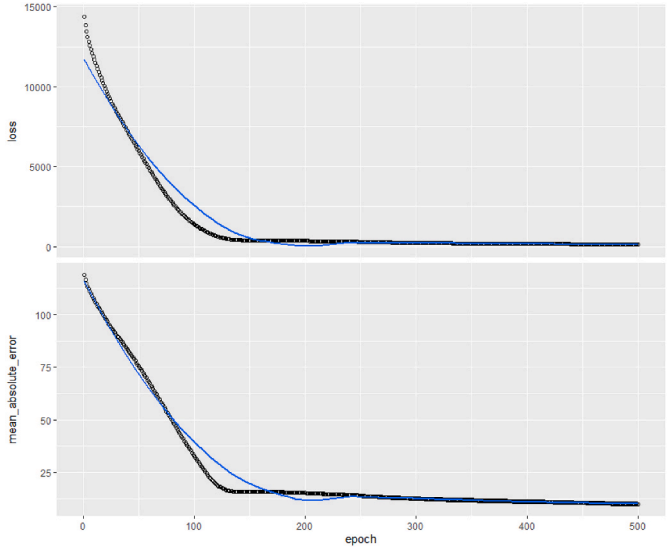


Fig. 12. Training and validation accuracy changes with the number of epochs using DNN.

The chart shows that pH has the highest influence with a value greater than 2, followed by turbidity (greater than 2), temperature (greater than 1.25), COD (greater than 1), BOD (1), DO (greater than 0.5) and nitrate, TDS, and EC have sensitivity values less than 0.5. The results indicate that major changes in WQI occur with minor changes in pH and turbidity and vice versa. To control WQI or reduce water pollution in Loktak Lake, pH and turbidity must be closely monitored and the sources of pH and turbidity pollution identified for control. Sensitivity analysis should be performed before the management process begins. The last density diagram illustrates the uncertainty analysis of the input parameters for the output WQI. The sensitivity analysis quantifies the impact of the input parameters on the output, while the uncertainty analysis highlights how uncertainties in the input parameters affect the output. It displays the distribution pattern of the input parameters and can reveal linear and non-linear relationships with the output. From the calculated mean and standard deviation of the sensitive value of each input parameter for the output, the coefficient of variation (CV) was calculated to show the variation in data distribution pattern. A higher variation indicates greater uncertainty in the output. In this study, EC showed high CV (44.15%), followed by TDS (negative 38%) and COD (23.75%). For the remaining parameters, CV is less than 10%. This indicates that EC and TDS have high uncertainty for the output, while the rest of the parameters have low uncertainty for the output. As a result, it can be concluded that TDS and EC can be excluded from the WQI modeling process for a better outcome.

4. Discussion

Water pollution is a critical environmental issue that can harm both the environment and human health (Zhu et al., 2022). Contaminants in water bodies can negatively impact aquatic life and the ecosystem, as well as potentially spreading waterborne diseases to humans (Yang et al., 2021). This underlines the significance of developing effective water pollution management strategies (Hmoud Al-Adhaileh and WaselallahAlsaade, 2021; Jiang et al., 2021).

This study aimed to predict the WQI using a stacking ensemble model, which combines GBM, GLM, and Neural Network models. The study also utilized DNN and CNN models, which were optimized using a grid search approach for improved performance in data-scarce conditions. This study is unique in that it is the first to implement a DNN model for sensitivity and uncertainty analysis in water pollution studies, which can aid decision making. Additionally, this comprehensive study marks the first time it has been conducted in the Loktak Lake. Previous research in the area has mainly focused on computing the WQI, prediction, geochemical analysis, and phumdis (Das et al., 2017; Roy and Majumder, 2019; Mayanglambam and Neelam, 2022; Laishram and Dey, 2014; Khwairakpam et al., 2020, 2021). This new research will provide a valuable contribution to reducing water pollution in the Loktak Lake by helping to implement cleaner production technologies.

The results showed that the computed WQI ranged from 90.75 to 145.29, which is classified as very poor to unsuitable according to the WQI classification scheme of Yadav et al. (2010). 70% of the sample sites were considered unsuitable for water quality, while 30% were considered very poor. Laishram et al. (2022) found that the WQI of the Loktak Lake varied from 38.19 to 155.47 during the pre-monsoon season and from 39.48 to 432.26 during the post-monsoon season. The lake is primarily polluted due to waste dumping and untreated sewage being dumped into it from multiple rivers. Waste water treatment techniques have been developed (Hu et al., 2022) that can reduce water pollution, so it's important to continue monitoring and implementing management initiatives for the protection of this Ramsar site.

The proposed stacking model outperformed other models in predicting WQI, while the CNN model also performed well with limited data. Stacking models are known to provide highly accurate predictions for WQI (Nasir et al., 2022) and have been shown to outperform other single and ensemble models in predicting not only WQI, but other issues

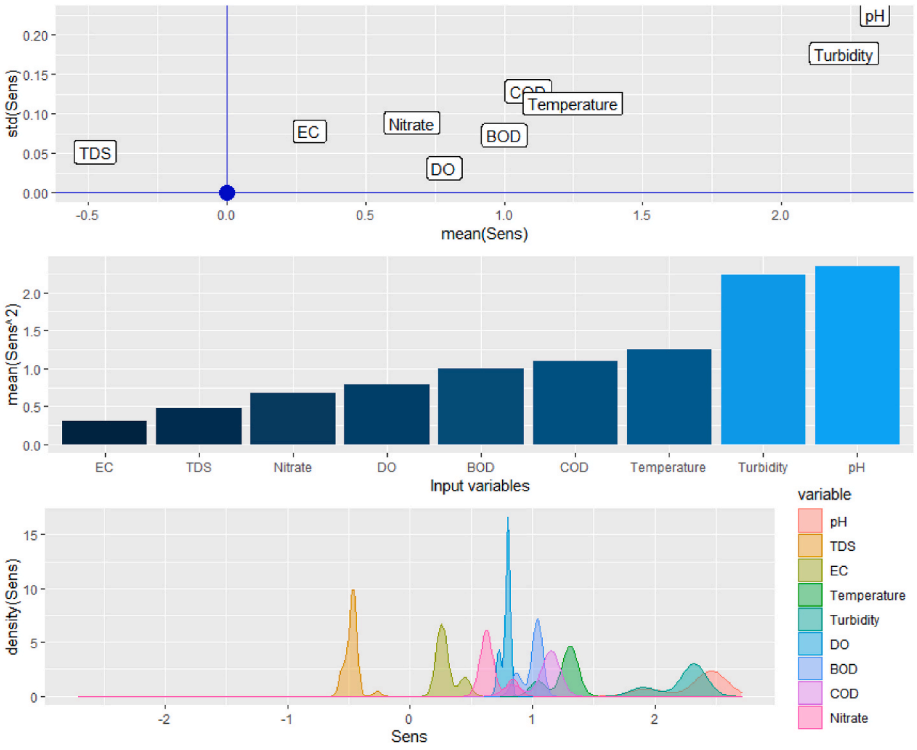


Fig. 13. Sensitivity and uncertainty analysis using deep neural network based sensitivity technique.

as well (Taghizadeh-Mehrjardi et al., 2020; Hu et al., 2020; Dou et al., 2020; Mallick et al., 2022). The reason for the superiority of stacking models is their ability to merge the strengths of strong models with meta-learners. In addition, this study suggests that deep learning models can perform well with limited data if they are fine-tuned. These models can provide valuable information on the important variables and their impact on WQI, which is crucial for effective water pollution management.

As per author's knowledge there is limited research exploring sensitivity and uncertainty analysis for decision-making, but common methods such as sobol indices, eFAST, FAST, and MORIS have been used with success (Talukdar et al., 2022; Mallick et al., 2022). This study chose to use a hyper-tuned DNN model to investigate its performance in predicting the WQI with limited data. The results show that the hyper-tuned DNN model is effective and the sensitivity matrices and uncertainty measures are reliable even with limited data, which could be useful in areas where data is scarce. The sensitivity analysis reveals that pH and turbidity are the most significant parameters affecting WQI and should be prioritized in monitoring and management efforts, which aligns with previous research (Jamal et al., 2022; Wu et al., 2018; Mohseni-Bandpei et al., 2018). The study also highlights the high level of uncertainty in EC and TDS, which is supported by previous research (Aldrees et al., 2022; Singha et al., 2021).

Based on the above discussion, it can be stated that the findings of this helps to promote sustainable development and improve environmental performance through implementing cleaner production practices. This study contributes to the field by adopting a comprehensive approach to water pollution management, including the use of advanced models for real-time monitoring and identification of major pollution contributors. The results provide valuable insights into the most significant factors affecting water quality and sources of uncertainty in prediction, which can inform industrial organizations and development agencies in their efforts to implement the cleaner production practices and better water quality management. The pioneering nature of this study highlights the importance of further research and development in this field, making it a valuable addition to the sustainable water pollution management.

5. Conclusion

The study shows that stacking ensemble model and optimized deep learning models i.e. DNN and CNN provide high accuracy in predicting WQI with limited data. The stacking ensemble model outperformed the other five models; CNN and DNN models performed well, nearly matching the performance of the stacking ensemble model. This highlights the potential for these optimized deep learning models to provide reliable water quality information even with limited data. The use of DNN for sensitivity analysis in this study is a novel and innovative approach in water pollution research because DNN has never been used for such analysis before. The results indicate that pH and turbidity are the most critical parameters affecting water quality, emphasizing the need for wastewater treatment, pollution prevention, liquid waste disposal, source reduction, and education and outreach programs. These cleaner production practices can help reduce the levels of contaminants in the water, leading to better water quality, human health and healthier ecosystems. The findings of this study can contribute to the development and adoption of cleaner production technologies in water pollution management. The study provides a low-cost and highly effective solution for sustainable water pollution management. This research may be particularly useful in regions with limited data availability and similar geographical conditions, where conventional water quality monitoring methods may be difficult or expensive to implement. Further studies can be directed towards application of IoT based electrochemical sensors for development of efficient early detection and warning system for water quality monitoring and management. As well a low cost framework can be developed using electrochemical microorganism-based bio sensors embedded with IoT platform to obtain real time monitoring of water quality of major lakes, water bodies.

CRediT authorship contribution statement

Swapan Talukdar: Conceptualization, Data curation, Formal

analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Shahfahad:** Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Shakeel Ahmed:** Writing – review & editing. **Mohd Waseem Naikoo:** Writing – review & editing. **Atiqur Rahman:** Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Santanu Mallik:** Resources, Writing – original draft, Writing – review & editing. **Sudhakar Ningthoujam:** Writing – review & editing. **Somnath Bera:** Writing – review & editing. **Gunturi Venkata Ramana:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2023.136885.

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